The global hydrogen budget

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Hydrogen (H_2) will play a part in decarbonizing the global energy system[1]. However, hydrogen interacts with methane, ozone, and stratospheric water vapour, leading to an indirect 100-year global warming potential of 11 ± 4 (refs. 2–5). This raises concerns about the climate consequences of increasing H_2 use under future hydrogen economies[3,5]. A comprehensive accounting of H_2 sources and sinks is essential for assessing changes and mitigating environmental risks. Here we analyse trends in global H_2 sources and sinks from 1990 to 2020 and construct a comprehensive budget for the decade 2010–2020. H_2 sources increased from 1990 to 2020, primarily because of the oxidation of methane and anthropogenic non-methane volatile organic compounds, biogenic nitrogen fixation, and leakage from H_2 production. Sinks also increased in response to rising atmospheric H_2. Estimated global H_2 sources and sinks averaged 69.9 ± 9.4 Tg yr^{-1} and 68.4 ± 18.1 Tg yr^{-1}, respectively, for 2010–2020. Regionally, Africa and South America contained the largest sources and sinks of H_2, whereas East Asia and North America contributed the most H_2 emissions from fossil fuel combustion. We estimate that rising atmospheric H_2 between 2010 and 2020 contributed to an increase in global surface air temperature (GSAT) of 0.02 ± 0.006 °C. GSAT impacts of changing atmospheric H_2 in future marker Shared Socioeconomic Pathway scenarios are estimated to remain within 0.01–0.05 °C, depending on H_2 usage, leakage rates and CH_4 emissions that influence photochemical H_2 production.

Hydrogen (H_2) has received increased attention as an energy carrier to help decarbonize heavy industry and transport and to provide long-duration energy storage[1]. When produced by electrolysis with renewable energy, hydrogen can, in principle, be produced and consumed with near-zero carbon emissions. As a result, many energy-system scenarios project substantial growth in H_2 production and utilization this century[1,6].

At present, hydrogen production is energy- and greenhouse gas-intensive. More than 90% of hydrogen produced today is grey hydrogen, derived mainly from steam methane reforming or coal gasification, which are both carbon-intensive[1]. In anticipated net-zero scenarios, however, a shift towards cleaner, low-carbon hydrogen production is projected[1,7]. This transition encompasses both green hydrogen produced through electrolysis powered by low- or zero-carbon electricity,

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and blue hydrogen generated by reforming fossil fuels coupled with carbon capture and storage. Ultimately, green and blue hydrogen are projected to dominate hydrogen manufacturing by 2030–2040 (ref. 6).

Excessive atmospheric H_2 has climate consequences. Despite its relatively short lifetime of 1.9–2.7 years (ref. 5), H_2 acts as an indirect greenhouse gas. By consuming OH radicals, a crucial sink for CH_4, H_2 warms the climate indirectly by extending the lifetime of CH_4, producing greenhouse gases such as ozone and stratospheric water vapour, and affecting the formation of aerosols and clouds[2,5]. Recent studies estimate 20-year and 100-year global warming potentials for H_2 of 37 ± 18 and 11 ± 4, respectively[2–5] (Supplementary Table 1). Understanding the impact of hydrogen leakage is, therefore, necessary to realize the full climate benefits of the H_2 economy[2,3,8–10]. For blue hydrogen, additional climate impacts arise from fugitive CH_4 emissions and uncaptured CO_2, which, in scenarios with relatively high H_2 and CH_4 emissions, can result in greater warming than directly burning natural gas[3,11].

H_2 concentrations in the atmosphere increased by about 70% from preindustrial times through 2003, after which time its concentration briefly stabilized[12,13]. However, H_2 concentrations began increasing again around 2010 (ref. 14) (Supplementary Note 1), reaching an annual mean level of around 555 ppb in 2024 (ref. 15). The reasons for this recent rise remain poorly quantified.

Here, we examine the trends of H_2 sources and sinks over the past three decades (1990–2020) and construct a comprehensive global budget for the most recent decade (2010–2020) with gridded sources and sinks (Extended Data Fig. 1). We use bottom-up approaches for estimating H_2 sources and sinks. These approaches combine activity data, H_2 emission factors, and emission factors or emission inventories of precursor species with process-based modelling. Using an ensemble of synthesized activity data, updated emission inventories, recently published emission factors and new observational measurements, we also provide detailed uncertainty estimates for key H_2 sources and sinks, which were poorly characterized previously.

With this comprehensive budget as a baseline for assessing future changes, we further estimate the climate impact of increasing H_2 concentrations historically and in future global H_2 economies.

Three decades (1990–2020) of global H_2 sources and sinks

The earliest global H_2 budget traces back to the 1980s (ref. 16), with several later updates[17–23]. None of these previous studies fully examined recent changes in the H_2 budget attributable to rapid changes in the emissions and processes of precursor gases, particularly CH_4 and non-methane volatile organic compounds (NMVOCs), or for OH burdens, H_2 concentrations and terrestrial and technological processes.

We estimate that increasing H_2 emissions from 1990 to 2020 arose predominantly from anthropogenic sources (that is, oxidation of increased levels of CH_4 and anthropogenic NMVOCs and leakage from H_2 production) (Fig. 1). By contrast, natural sources such as fire emissions and biogenic NMVOC oxidation show substantial interannual variation but no notable trends over this period (Fig. 1). The global surface average H_2 concentration increased from 523.4 ppb in 1992 to 543.5 ppb in 2020, a 3.8% increase, consistent with the increasing anthropogenic emissions of CH_4 and NMVOCs during the study period.

We estimate H_2 production from CH_4 oxidation to be the largest increasing source from 1990 to 2020 (0.1 Tg yr^{-1} or a total increase of about 4 Tg), attributable mainly to increasing atmospheric CH_4 from anthropogenic activities[24], but there was also a sudden decrease in 2020 attributable to COVID-19 (Fig. 1). H_2 leakage from H_2 production increased at an estimated rate of 0.015 Tg yr^{-1} because of an increasing industrial usage of H_2 (that is, a total increase of around 0.45 Tg from 1990 to 2020) (Fig. 1). H_2 emissions from the oxidation of anthropogenic NMVOCs and biogenic nitrogen fixation (BNF) increased at an estimated rate of 0.005 Tg yr^{-1} and 0.008 Tg yr^{-1}, respectively (Fig. 1). H_2 is produced naturally from BNF, but the increase is probably attributable to increased anthropogenic cultivation of leguminous crops[25]. Some anthropogenic sources of H_2 seem to be decreasing, such as direct emissions from the combustion of fossil fuels and biofuels (estimated declines of −0.1 Tg H_2 yr^{-1} and −0.03 Tg H_2 yr^{-1}, respectively), most likely attributable to improvements in combustion efficiency in engines that reduce incomplete combustion[26]. Nevertheless, we acknowledge that the H_2/CO emission ratio may have evolved over time as technology improves, but a constant H_2/CO ratio was assumed in this study.

Both the global soil and OH sinks increased significantly over the 1992–2020 period at rates of 0.05 Tg yr^{-1} and 0.06 Tg yr^{-1} ($P \ll 0.01$), respectively, attributable to increasing H_2 in the atmosphere (Fig. 1). We also estimate a small but significant increase in the global average soil H_2 deposit velocity (slope = 7×10^{-6} cm s^{-1} yr^{-1}, $P < 0.05$) in 1992–2020.

The global H_2 budget (2010–2020)

We estimate the global H_2 budget for the recent decade (2010–2020) by synthesizing multiple datasets and models, incorporating uncertainties propagated from multiple levels, including activity data, precursor emissions, and emission factors for both precursor species and H_2 (Methods). No previous analysis, to our knowledge, estimated the uncertainties of these sink/source terms by synthesizing multiple datasets or modelling analyses as done here.

We estimate mean global H_2 sources and sinks to be 69.9 ± 9.4 Tg yr^{-1} and 68.4 ± 18.1 Tg yr^{-1}, respectively, for the decade 2010–2020 (Fig. 2). The largest anthropogenic sources of H_2 over this decade include those from fossil fuel and biofuel combustion, leakage from industrial H_2 production, and oxidation of CH_4 and anthropogenic NMVOCs (Fig. 2). Photochemical oxidation is the largest H_2 source (an estimated 38.4 Tg yr^{-1}, 56% of the global total), whereas soil uptake is the largest sink (an estimated 50.0 Tg yr^{-1}, 73% of total sinks) (Fig. 2). Soil uptake contributes the largest uncertainty to our total budget uncertainty, followed by photochemical oxidation, fossil fuel combustion, BNF, and biomass and biofuel burning.

The H_2 budget was further evaluated using a two-box model (Supplementary Note 2) to optimize total sources or total sinks for 2010–2020. Our bottom-up imbalance based on mean total sinks and sources is 1.5 ± 20.4 Tg yr^{-1}, whereas the optimized imbalance is 0.6 ± 1.4 Tg yr^{-1} at a mean atmospheric lifetime for H_2 of 2.8 years (Extended Data Table 1). The mean difference between previous and optimized values is less than 1 Tg yr^{-1}, within the uncertainty range of the priors, indicating that only a modest adjustment to either emissions or losses can considerably improve agreement with observed atmospheric H_2. Our estimated balance between emissions and losses also closely matches the interannual burden change ($R^2 = 0.42$).

Our global total source estimate falls within the range of previous estimates for earlier time periods (Supplementary Table 2). However, it is approximately 30 Tg smaller than the two previous atmospheric inversion studies[18,20], which probably overestimated H_2 production from NMVOCs (>36 Tg yr^{-1}) for the following reasons. These higher estimates conflict with the well-established NMVOC production levels of carbon monoxide (CO), whose primary removal mechanism (reaction with OH, about 90%) is widely recognized to limit its sources. Moreover, satellite observation of atmospheric formaldehyde (HCHO), which is produced from NMVOCs to generate H_2, does not support such a high level of NMVOC production of H_2 (Methods).

Our estimate of total sinks agrees well with most previous studies[17,21,22,27–30]. We estimated OH oxidation and its uncertainties based on an ensemble of eight OH datasets and the soil sink and its uncertainty based on seven different model parameterization and 10 different soil attribution inputs (Methods).

The spatial distribution of sources is more uneven than for sinks (Fig. 3). Hotspots of H_2 emissions are evident, with the highest density of emissions observed in Southeast and East Asia. However, tropical

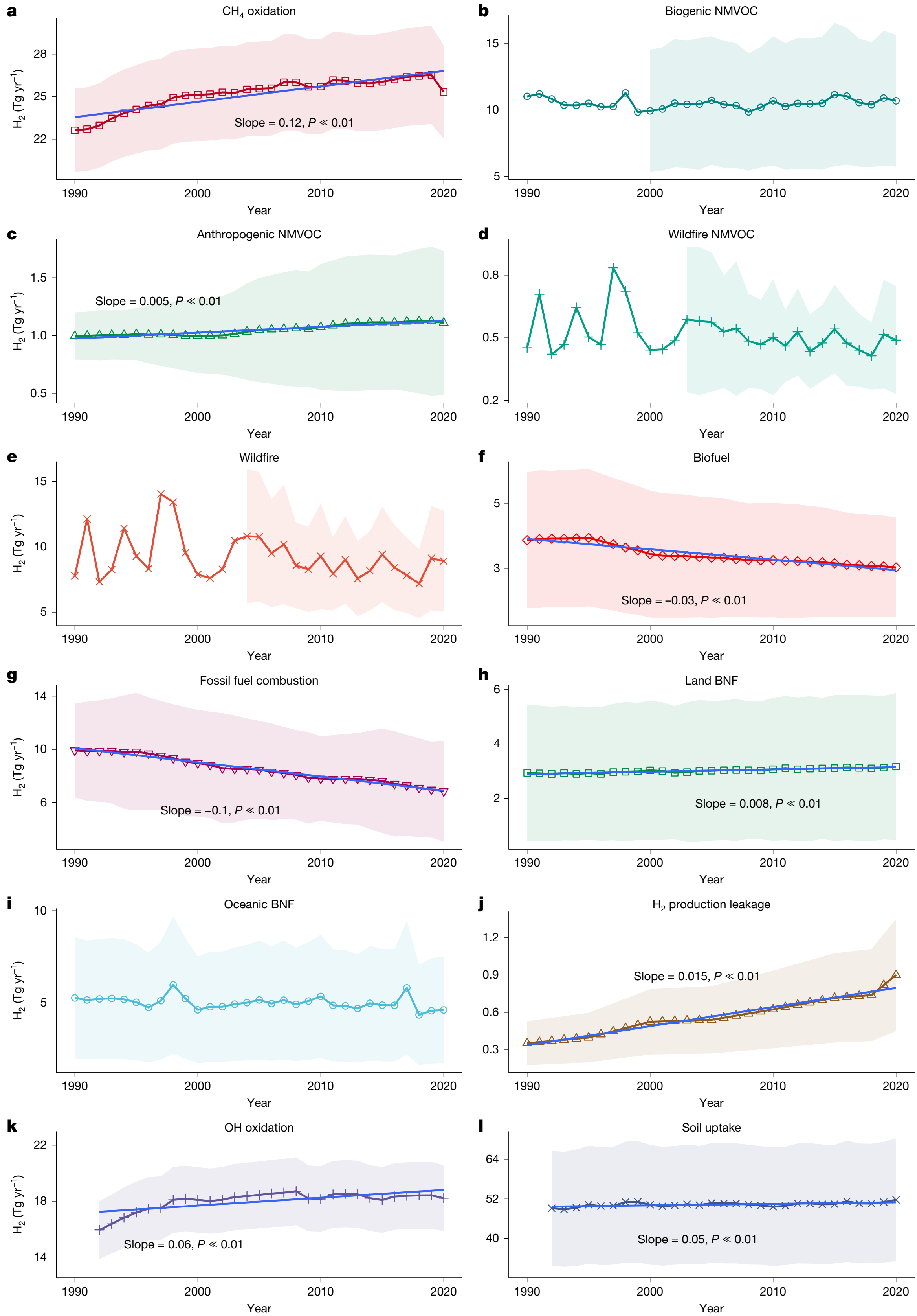

Fig. 1 | Global sources and sinks of H_2 (Tg H_2 yr^{-1}) over three decades (1990–2020). a–j, The H_2 sources include production from oxidation of CH_4 (**a**), biogenic NMVOCs (**b**), anthropogenic NMVOCs (**c**) and wildfire NMVOCs (**d**); as well as direct H_2 emissions from wildfires (**e**), combustion of biofuels (**f**), the combustion of fossil fuels (**g**), biological nitrogen fixation on land (**h**) or in the oceans (**i**), and H_2 leakage during industrial H_2 production (**j**). **k**,**l**, The sinks (bottom two panels) include H_2 oxidized by OH (**k**) and H_2 uptake by microbes in soils (**l**). The category 'Anthropogenic NMVOC' (**c**) includes emissions of NMVOCs from burning both fossil fuels and biofuels. Note that some minor sources shown in Fig. 2 are omitted for this time series analysis. Uncertainties (shaded regions) represent standard deviations (1 s.d. above and below the line).

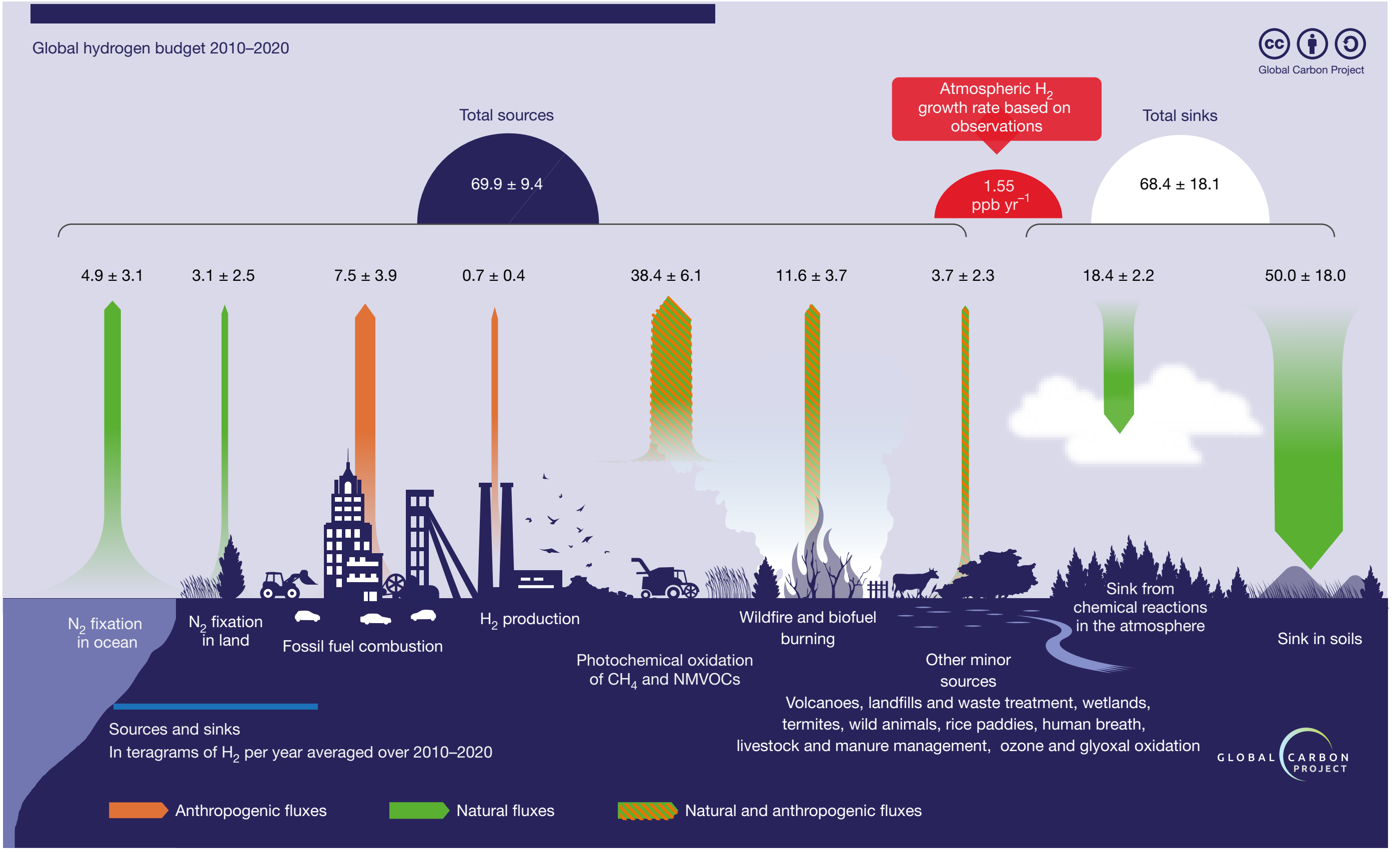

Fig. 2 | The main sources and sinks of H_2. The main sources (upward arrows) and sinks (downward arrows) of H_2 (Tg yr^{-1}) for anthropogenic (orange), natural (green) and mixed anthropogenic and natural fluxes (hatched). Dominant sources include photochemical oxidation of both CH_4 and non-methane volatile organic compounds (NMVOCs), combustion of biomass and fossil fuels and H_2 production from nitrogen fixation. The dominant global sinks of H_2 are consumption by soil microbes and oxidation by tropospheric hydroxyl radicals (OH).

regions contribute the largest share (about 60%) of the total amount of emissions and production (Fig. 3a). This result is attributable to the combination of higher temperatures in the tropics promoting CH_4 and NMVOC oxidation, abundant plant biomass that leads to relatively high biogenic NMVOC emissions, and frequent tropical fires. Although the distribution of H_2 sinks is more uniform across non-desert and non-frozen lands globally, tropical regions still account for the largest sink (around 50% of the global total) (Fig. 3b).

Photochemical H_2 production

H_2 is produced in the atmosphere through the photolysis of formaldehyde (HCHO) (ref. 17). HCHO is produced by the oxidation of CH_4 and NMVOCs by OH, with yields affected by levels of NO_x gases[31]. Previous estimates of production from CH_4 oxidation range from 15 Tg yr^{-1} to 27 Tg yr^{-1} (refs. 21,23,28), but earlier studies either did not report uncertainties or disclosed much larger uncertainties (≥±8 Tg yr^{-1}) (refs. 17,21) than ours (± 3.5 Tg yr^{-1}). We estimate a relatively higher average rate of 26.1 ± 3.5 Tg H_2 yr^{-1} for this source (Supplementary Table 2), primarily attributable to increasing atmospheric CH_4.

Various NMVOCs, including biogenic sources, anthropogenic sources and mixed sources such as biomass burning, produce atmospheric H_2 through photochemical oxidation. We estimate that biogenic, wildfire and anthropogenic NMVOCs contribute 10.7 ± 5.0 Tg yr^{-1}, 0.5 ± 0.3 Tg yr^{-1} and 1.1 ± 0.6 Tg yr^{-1}, respectively (Extended Data Table 1), summing to 12.3 ± 5.0 Tg yr^{-1}.

Combining CH_4 and NMVOCs oxidation, we estimate total photochemical sources of H_2 to be 38.4 ± 6.1 Tg yr^{-1} for the decade 2010–2020. This quantity lies within the range of previous estimates (31–87 Tg yr^{-1}) and aligns well with a few estimates in other periods using three-dimensional (3D) chemical transport models[18,22,30] (Supplementary Table 2).

Direct H_2 emissions from combustion

Incomplete combustion generates CO, which can undergo the water–gas shift reaction to produce H_2 (ref. 17). Automobile transportation is the primary H_2 source from fossil fuel combustion, although other processes involving fossil fuel or biomass combustion also emit H_2.

Based on CO emissions and CO/H_2 emission ratios across various sectors of fossil fuel and biomass combustion (Methods), we estimate total H_2 emissions from fossil fuel combustion to be 7.5 ± 3.9 Tg yr^{-1} (3.7 ± 2.0 Tg yr^{-1} for automobile transport and 3.8 ± 3.3 Tg yr^{-1} from other fossil sources) and from biomass burning to be 11.6 ± 3.7 Tg yr^{-1} (8.4 ± 3.3 Tg yr^{-1} for wildfires and 3.2 ± 1.7 Tg yr^{-1} for biofuels) during 2010–2020. Our estimate of H_2 emissions from fossil fuel combustion is lower than that in previous studies (Supplementary Table 2). Our estimate of H_2 emissions from biomass burning (11.6 ± 3.7 Tg yr^{-1}) falls within the range of previous estimates[17,18,21,28–30,32].

Leakage from H_2 production

Similar to methane leakage from the natural gas industry, the production, distribution and utilization of H_2 have some unavoidable leakage. At present, however, the vast majority of H_2 production (>99%) is consumed locally at the site of production for industrial processes[1]. Estimates of current H_2 leakage, therefore, mainly consider leakage at production sites. The average global leakage rate for H_2 production is

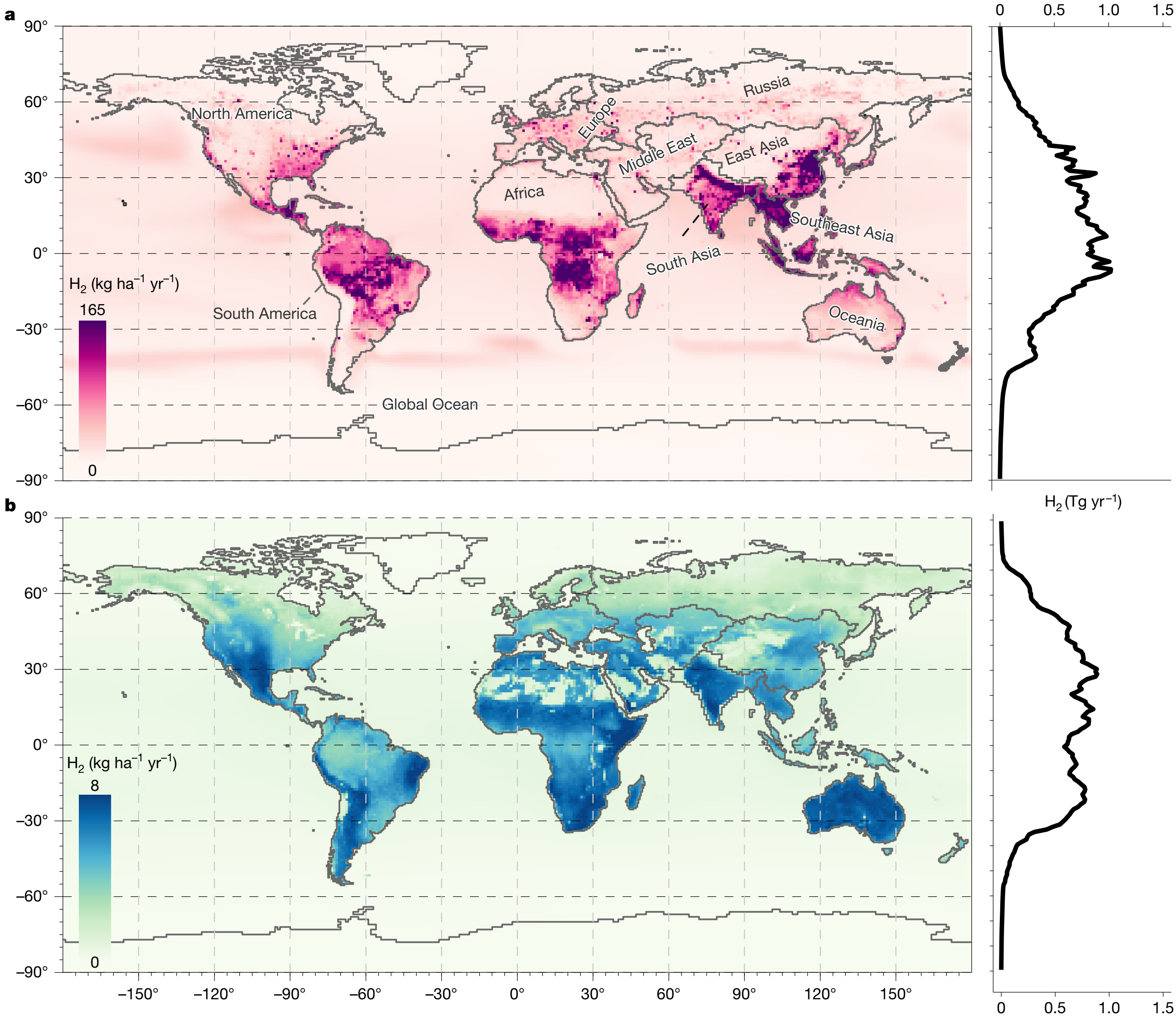

Fig. 3 | The spatial distribution of total sources and sinks of hydrogen. **a**, Total sources gridded spatially (top left) and the latitudinal distribution of sources (top right). **b**, Total sinks gridded spatially (bottom left) and latitudinally (bottom right). The 10 land regions of Earth follow the categorizations in the second phase of the REgional Carbon Cycle Assessment and Processes Project (RECCAP-2)[59] and are labelled in **a**. These regions are used subsequently to analyse regional sources and sinks. The total sources here include only the photochemical oxidation of CH_4 and NMVOCs, direct emissions from fires and the combustion of fossil fuels and biofuels, and emissions from biological nitrogen fixation terrestrially and in the oceans.

relatively unconstrained because of a lack of measurements. Using a 1 ± 0.5% leakage rate for industrial H_2 production (Methods), we estimate total H_2 leakage at 0.7 ± 0.4 Tg yr^{-1} for 2010–2020 (Fig. 1), within the range of previous estimates[23,33].

H_2 emission from BNF

H_2 is produced as a by-product of BNF[34], a widespread process carried out by symbiotic and free-living bacteria. However, the H_2 sources from oceanic and terrestrial BNF remain relatively poorly constrained. Previous estimates suggest a range of 0–11 Tg yr^{-1} from land and oceans combined (Supplementary Table 2).

Some studies ignored land-based sources of H_2 from BNF, arguing that most soil H_2 production is consumed by soil hydrogenases[29]. However, observations in various land covers show H_2 escaping from the soil to the atmosphere[34,35], indicating the importance of this source. Through an ensemble of models that estimated N fixation from BNF and newly compiled H_2/N_2 net production ratios (Methods), we estimate a new land-based H_2 emission to be 3.1 ± 2.5 Tg yr^{-1} for the period 2010–2020 (Fig. 1).

Previous studies observed H_2 supersaturation in oceans with respect to atmospheric H_2 (refs. 36–39), leading to H_2 release into the atmosphere. Oceanic H_2 mainly originates from cyanobacterial BNF, although other origins are possible[17,40]. Using our newly compiled H_2/N_2 net production ratios from the literature (Methods), we estimate the total oceanic BNF H_2 emission to be 4.9 ± 3.1 Tg yr^{-1} for 2010–2020 (Fig. 1).

Other minor H_2 sources

There are additional minor H_2 sources (usually <1.0 Tg H_2 yr^{-1}) with substantial uncertainties. Our budget accounts for one geological source (volcanic emissions) and eight fermentation processes (the breakdown of organic materials under anaerobic conditions) occurring in wetlands, rice paddies, landfills, waste treatment, livestock production (that is, enteric fermentation and manure management), termites, wild animals and human breath, along with two photochemical processes (the photolysis of ozone and glyoxal). Collectively, we estimate a total of 3.7 ± 2.3 Tg H_2 yr^{-1} from these minor sources (Fig. 2) based on sparse data of emission factors compiled from the literature (Methods). Although individually small compared with major sources, their cumulative emissions nevertheless represent a substantial contribution to atmospheric H_2 and warrant further study and elucidation.

H_2 sinks

Previous estimates of global soil H_2 uptake ranged widely from 15 Tg yr^{-1} to 99 Tg yr^{-1} (Supplementary Table 2). We used the process-based model formulation in ref. 41 to estimate global soil H_2 uptake with seven sets of parameterizations. Soil properties are recognized as a large source of uncertainty[42]. We thus also integrated forcing data for soil texture, porosity, temperature, moisture and snow depth from an ensemble of 10 global models to estimate this uncertainty. We estimate a global H_2 soil uptake rate of 50.0 ± 18.0 Tg yr^{-1} for the period 2010–2020 (Fig. 2), a range that includes most previous estimates (Supplementary Table 2).

The loss of H_2 through reactions with OH in the atmosphere is relatively well-constrained compared with soil uptake. However, it carries uncertainty attributable in part to variations in OH distributions and trends. By considering the 3D distribution of OH and H_2, along with temperature-dependent reaction rates[43], we estimate this loss of H_2 to be 18.4 ± 2.2 Tg yr^{-1} during the decade 2010–2020. Our estimate is also consistent with most previous studies[17,18,20–22,30,32,44] (Supplementary Table 2).

Regional sources and sinks (2010–2020)

Our estimates show strong regional differences in the magnitude of H_2 sources and sinks (Fig. 4). Most H_2 photochemical production/loss occurs over the oceans, because of their large global area, particularly in the tropics, in which both OH concentrations and temperatures are relatively high. On land, Europe is the smallest regional source of H_2 (<2% of global total), whereas Africa and South America stand out as the largest two regional sources (about 16% and 11% of the global total, respectively) (Supplementary Table 3). Africa is also the largest regional sink for H_2 (about 23% of the global total), followed by South America as the second largest (around 13% of global total). Southeast Asia is the smallest regional sink (approximately 4% of the global total) (Supplementary Table 3).

The two largest economic regions of the world, East Asia and North America, contribute the most H_2 emissions from fossil fuel combustion (about 32% and 15% of the global total, respectively; Supplementary Table 3). South America is the largest contributor of H_2 from land-based BNF (27% of the terrestrial total). This result can be attributed to its extensive cultivation of leguminous plants such as soybeans and peanuts[45], as well as extensive tropical areas in which native species fix nitrogen[46].

H_2 generated from biomass combustion, NMVOCs and CH_4 oxidation comprise the dominant sources in Africa and South America (Fig. 4). In Africa, biomass combustion, mostly through wildfires, constitutes the largest source, followed by NMVOC oxidation (Fig. 4). In South America, NMVOC oxidation is the largest source, followed by biomass combustion (Fig. 4). Both Africa and South America have extensive tropical areas and are characterized by frequent wildfires[47] and abundant vegetation that produces NMVOCs[48]. H_2 emissions from fossil fuel combustion comprise the largest source of H_2 in East Asia, North America and Europe (Fig. 4), consistent with more intensive fossil fuel use there.

Climate impacts of atmospheric H_2

We estimate changes in atmospheric H_2 and their associated impact on GSAT using the compact Earth system model OSCAR[49], which integrates the hydrogen cycle and its interactions with methane (CH_4) (Methods). Unlike previous studies that primarily assess the net climate benefits of H_2 by examining tradeoffs between reductions in greenhouse gas emissions and potential H_2 leakage[3,50], our work estimates changes in the H_2 budget and quantifies the missing warming in the current climate simulations due to the lack of an interactive representation of the H_2–CH_4 system in most models.

Over the past decade (2010–2020), we estimate that rising atmospheric H_2 concentrations have contributed to an increase in GSAT of 0.020 ± 0.006 °C with respect to the preindustrial period. For the future, we project all components of the H_2 budget using IPCC AR6 marker Shared Socioeconomic Pathway (SSP) climate scenarios[51] to estimate the changes in atmospheric H_2 and the resulting climate impacts not currently included in these SSP climate projections.

The future contribution of H_2 to GSAT depends on how much H_2 is consumed as an energy carrier (Fig. 5a), leaks during usage (Fig. 5b) and is produced through the photolysis of CH_4 in the atmosphere (Fig. 5e, as photolysis of NMVOCs is not projected to change much) and other H_2 emissions (Fig. 5c,d). In low-warming scenarios with high H_2 usage (for example, SSP1-1.9 and SSP1-2.6; Fig. 5a), methane emissions are substantially mitigated, which reduces the formation of H_2 from CH_4 by photolysis (Fig. 5e). Therefore, the warming from H_2 could even decrease slightly from the present-day level if H_2 leakage is relatively low (for example, 1% leakage, Fig. 5g) but still increase if leakage is high (for example, 10% leakage, Fig. 5h). In a medium-warming scenario such as SSP2-4.5, methane emissions decline only slightly, and the change in warming from H_2 is predominately determined by the H_2 leakage rate: it may be similar to today under low leakage rates but increase substantially under higher leakage (Fig. 5g,h) (acknowledging that such a high H_2 usage is specific to this SSP2-4.5 (ref. 52) of the SSPs we examined; Fig. 5a). In higher-warming scenarios (for example, SSP 4-6.0 or SSP5-8.5), the H_2 use is relatively low but methane emissions remain largely unmitigated. In these cases, the additional H_2 formed from CH_4 photolysis can outweigh the H_2 that leaks from the system under both low and high H_2 leakage scenarios (Fig. 5g,h), increasing H_2-induced GSAT. Overall, this compensation effect caused by inversely related CH_4 and H_2 emissions across these limited SSP scenarios leads to a missing warming in climate projections of 0.01–0.05 °C, which is relatively small compared with the long-term total temperature decrease in mitigation, that is, about 1.0 °C in SSP2-4.5 (current policies), about 1.9 °C in SSP1-2.6 and about 2.5 °C in SSP1-1.9 compared with an SSP3-7.0 (baseline)[51]. However, if there were scenarios with both high demand and leakage rates for H_2, as well as unmitigated CH_4 emissions, the potential warming resulting from increasing H_2 concentrations would be considerably larger than those shown in Fig. 5.

Discussion

Our new global H_2 budget for the recent decade 2010–2020 was estimated by synthesizing multiple datasets and models, with each budget term estimated individually using a bottom-up synthesis approach. When combined with atmospheric H_2 observation data[15], our gridded H_2 sources and sinks can provide data needed for further optimization of the global H_2 budget by top-down atmospheric inversions and bottom-up chemistry transport models.

Our work also highlights remaining uncertainties in sources and sinks, and research needs to constrain the global H_2 budget further. The largest uncertainty arises from the soil uptake term, which is sensitive to both model parameterization (especially the value of the maximum biological uptake rate) and intermodel variation of soil attributes. Refined quantification of soil characteristics (for example,

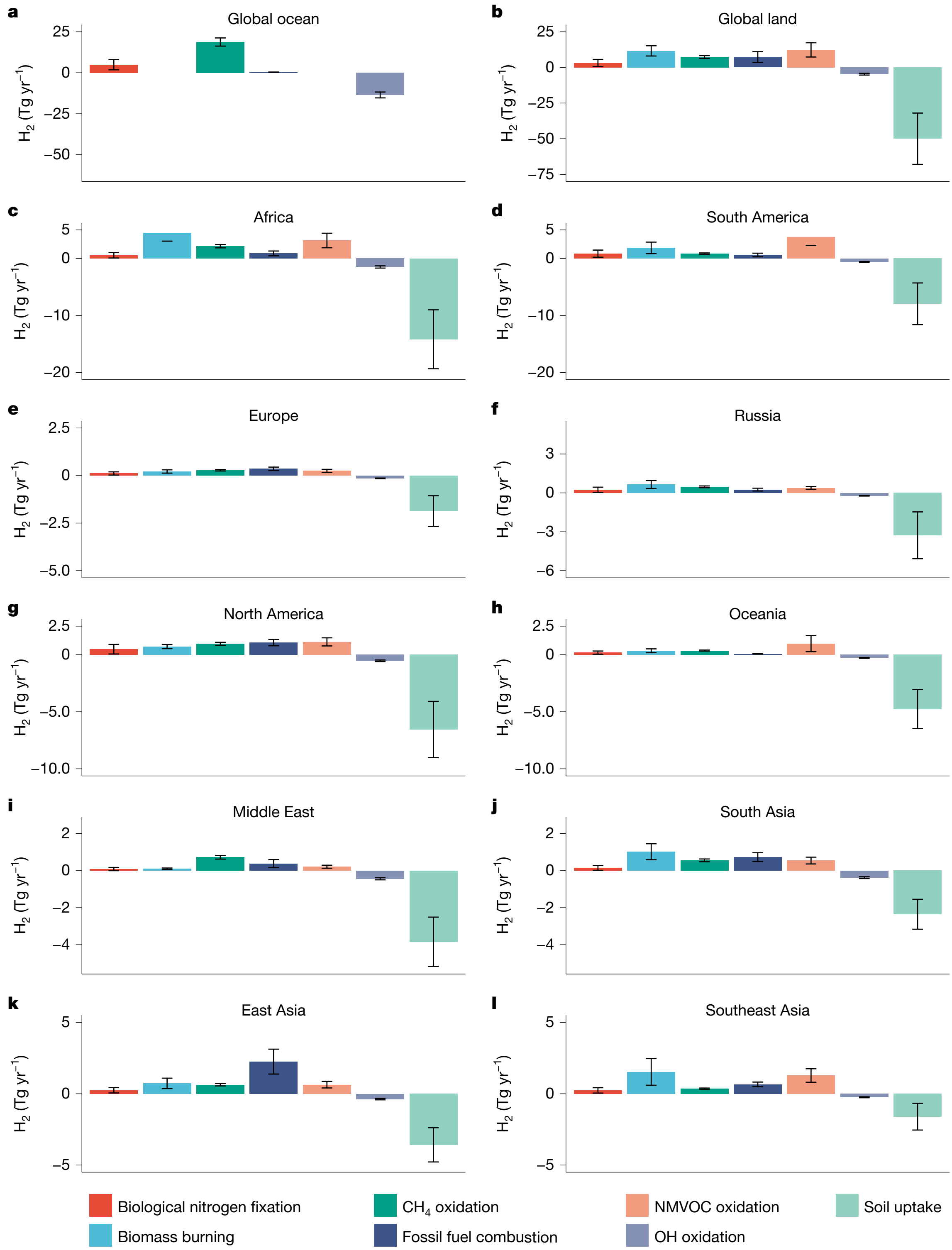

Fig. 4 | Regional sources and sinks of H_2 (Tg H_2 yr^{-1}) averaged for the decade 2010–2020. a,**b**, The Earth is partitioned into global ocean (**a**) and global land (**b**). **c**–**l**, The ice-free land of Earth is further partitioned into 10 regions following the definitions used in the second phase of the Regional Carbon Cycle Assessment and Processes Project (RECCAP-2)[59]: Africa (**c**), South America (**d**), Europe (**e**), Russia (**f**), North America (**g**), Oceania (**h**), Middle East (**i**), South Asia (**j**), East Asia (**k**) and Southeast Asia (**l**). Each subplot shows the emissions from five source terms (shown as positive values) and two sink terms (shown as negative values). Minor source terms and H_2 leakage (Fig. 2) are not shown here.

soil porosity, composition, moisture and temperature) and understanding of how they influence microbial hydrogenases are needed to improve modelling of soil H_2 uptake. More in situ measurements of soil uptake rates are needed in different ecosystems and seasons to better validate, parameterize and constrain process-based H_2 uptake models. Moreover, more laboratory studies are needed to constrain the maximum biological H_2 uptake in different soil types and microbial communities.

Uncertainties associated with other sources and sinks are smaller than for the soil sink in absolute terms but can still be larger in relative

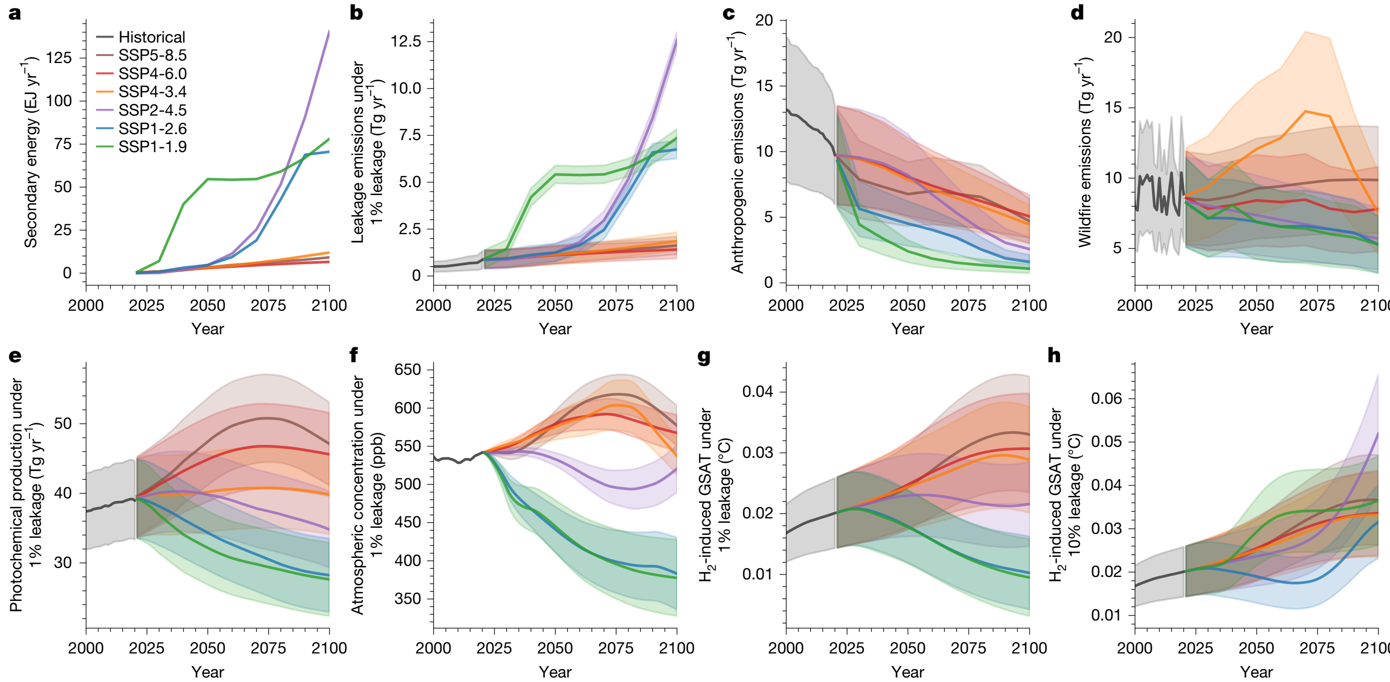

Fig. 5 | Projected global hydrogen use (EJ yr^{-1}), emissions (Tg yr^{-1}), atmospheric concentration (ppb), and climate impacts (°C) under varying leakage scenarios in marker SSP scenarios as used in IPCC AR6. a, Hydrogen as secondary energy in SSP scenarios through 2100. **b**, Corresponding leakage emissions by scenario under 1% H_2 leakage rate. **c**, Future anthropogenic emissions (fossil, biofuel and other minor sources). **d**, Future wildfire emissions. **e**, Total H_2 production from CH_4 and VOCs oxidation under 1% leakage rate. **f**, Atmospheric concentration of H_2 under 1% leakage rate. **g**,**h**, GSAT impact from H_2 concentration change under 1% (**g**) and 10% (**h**) leakage rates. The shaded regions represent 1 s.d. GSAT impact for other intermediate leakage rates (3% and 5%) can be found in Extended Data Fig. 2.

terms, and future work is needed to reduce these uncertainties. Increased H_2 production from CH_4 oxidation is probably the main reason for the increasing H_2 concentration in the atmosphere over the past decade, a conclusion that would benefit from further observations and studies to verify. The atmospheric distribution of CH_4 and the H_2-yielding mechanism of CH_4 oxidation are now fairly well understood, leading to a reduced uncertainty in H_2 source estimates from this process. However, the distribution of NMVOC and the molar yields of H_2 from NMVOC oxidation need further quantification across space and time. Other remaining uncertainties include estimates related to OH fields, particularly as the abundance of methyl chloroform declines below 1 pptv (parts per trillion by volume). Minimizing this uncertainty could further improve estimated photochemical production and loss of H_2.

There remains a substantial lack of empirical data on H_2 emission factors for various sectors, including minor sources, and more measurements are needed to improve the estimate to better constrain the global budget. More data are needed on leakage rates of H_2 in production sites, transport pipelines, and end-use facilities and appliances. Limited data also exist for H_2 emission factors for fossil fuel and biofuel combustion, nitrogen fixation processes on land and in the oceans, and for minor sources such as emissions from geological sources, wet soils (for example, wetlands, landfills and rice paddies) and enteric fermentation (for example, termites, and livestock), and photolysis of ozone and glyoxal. We summarize current data availability and gaps in Extended Data Table 2.

This lack of empirical emission data is caused largely by a lack of mobile or portable instruments that can quantify H_2 concentrations in air. For methane, for instance, laser-based instruments are available that measure CH_4 at low concentrations with relatively high precision and fast response. These instruments are used to quantify emission fluxes in chamber measurements, eddy covariance and tower monitoring, and in mobile and aerial measurements. The only commercially available H_2 instrument of comparable performance that we know of is the TILDAS H_2 Monitor manufactured by Aerodyne since 2024. This instrument was tested recently with controlled-release experiments to quantify H_2 emissions using a mobile platform[53]. However, to our knowledge, no H_2 monitors are as yet available for aerial, satellite and long-term tower-based monitoring.

Further work is needed to estimate sources and sinks that are potentially unaccounted for, including geologic seeps, pipeline leakage and vegetation. For example, we had insufficient data to consider geologic sources from geothermal seeps, hot springs and oceanic crusts beyond volcanoes. Moreover, there is limited data on H_2 leakage from natural gas systems, despite some commercial natural gas already containing blended H_2 (refs. 54,55).

Plant measurements are also needed because vegetation can be both a sink and a source of H_2. A Harvard Forest study observed unexpected aboveground emissions of H_2 during leaf senescence[56], although the mechanism of this emission was unclear. It may be a secondary emission, perhaps the oxidation of VOCs released by plants that are already accounted for in our current H_2 budget. However, if it is direct emission from plants, this could be a new source not included at present in H_2 budgets. Vegetation may also be an H_2 sink, as some research suggests that tree foliage and bark exhibit the capacity to absorb CH_4 (refs. 57,58). It is therefore possible, perhaps likely, that H_2-producing and H_2-consuming microbes coexist, as they do for CH_4.

We found that rising atmospheric H_2 between 2010 and 2020 contributed to an increase in GSAT of 0.020 ± 0.006 °C. Future climate impact could either decrease or increase depending on H_2 usage and leakage rates and CH_4 emissions but is expected to remain within 0.01–0.05 °C under IPCC marker SSP scenarios. Our results underscore the need for a deeper scientific understanding of the global hydrogen cycle and its links to radiative forcing to support a climate-safe and sustainable hydrogen economy.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-025-09806-1.

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